INCH-POUND
MIL-DTL-32122C (OS)
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SUPERCEDING
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29 March 2005

DETAIL SPECIFICATION

CARTRIDGE, AIRCRAFT FIRE EXTINGUISER JL42, JL43, JL44

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Department and Agencies of the Department of Defense.

1.0 SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the JL42, JL43 and JL44, Aircraft Fire Extinguisher Cartridge referred to herein as "cartridge".

1.2 Application. These cartridges are used to rupture one of the fire extinguisher tank's frangible disks allowing fire extinguishing agent to be rapidly discharged into the appropriate engine bay to put out the fire.

2.0 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202 Test Method Standard Electronic and Electrical Components Parts

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (E143D), 101 Strauss Avenue, Indian Head, MD 20640 5035, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this information using the ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 1377

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MIL-STD-1168 Ammunition Lot Numbering and Ammunition Data Card

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-23659 Initiators, Electric, General Design Specification For

(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

838AS185 Test Fixture, Cartridge Aircraft Fire Extinguisher

3205AS277-1 Cartridge, Cartridge Actuated JL42

3205AS277-2 Cartridge, Cartridge Actuated JL43

3205AS277-3 Cartridge, Cartridge Actuated JL44

(Application for copies should be addressed to the Commanding Officer, Naval Inventory Control Point, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.3 Non-Government publications. The following document form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

Pacific Scientific Energetic Materials Co.

2-103020-1, -2 and -3 Aircraft, Fire Extinguisher Cartridge

(Application for copies of Pacific Scientific Energetic Materials Co. drawings should be sent to Pacific Scientific Energetic Materials Co. ,7073 W. Willis Drive, Chandler, AZ 85226)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 748 Standard Practices for Thermal Neutron Radiography of Materials

ASTM E1742/E1742M Standard Practice for Radiographic Inspection

(Copies of this document are available online at http://www.astm.org/ or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AS8660 Silicone Compound NATO Code Number S-736

(Copies of this document are available online at http://www.sae.org/, or from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pennsylvania, United States, 15096-001.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3.0 REQUIREMENTS

3.1 First article. When specified by the procuring activity (see 6.3), a sample sufficient to perform all required tests shall be subjected to the first article inspection (see 4.2). The sample shall be manufactured using the same methods and procedures proposed for the production lot and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract. A first article inspection shall be required if;

a. Two or more years have passed since the cartridges were manufactured,
b. There has been a major turnover of personnel,
c. There have been major changes to the production facility or,
d. There has been a transfer of production to a new facility or site.

3.2 Conformance to documents. The Fire Extinguisher cartridges' DODICs: JL42, JL43 and JL44, covered by this specification shall be manufactured in accordance with drawing 3205AS277.

3.3 Design and construction. The aircraft fire extinguisher cartridge design and construction shall be identical to that subjected to qualification testing and approved for service use by the Navy. Any design changes affecting the interchangeability or performance of these items shall be subject to approval by the cognizant governmental design activity.

3.3.1 Configuration control. Only the cartridges procured from the vendor(s) listed on drawing 3205AS277 are approved for use by the Navy. The applicable DODIC and part numbers are shown in Table I. A substitute cartridge shall not be used or procured without prior approval and testing by the Indian Head Division, Naval Surface Warfare Center, and the Naval Air Systems Command Headquarters.

TABLE I. Configuration identification

DODIC	NAVAIR P/N
JL42	3205AS277-1
JL43	3205AS277-2
JL44	3205AS277-3

3.4 Cartridge inspection. The cartridges manufactured under contract shall be inspected and screened for defects as outlined in 3.4.1 through 3.4.6; failure of any cartridge to meet the requirements shall result in rejection of the cartridge.

3.4.1 Visual inspection. Each cartridge shall be rejected which has illegible, missing or inaccurate identification markings, damage or insufficient surface coatings, burrs, sharp edges, corrosion, dents,

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excessive pits or bubbles that may form a conductive path across the epoxy insulation surface, or other defects that might prevent proper installation or performance, when examined in accordance with 4.4.1.

3.4.2 Radiographic examination. Each cartridge pyrotechnics shall be properly oriented and without voids or discontinuities. All other parts shall be properly placed in accordance with manufacturers' drawings and in accordance with criteria listed below when inspected in accordance with 4.4.2.

Radiographic Acceptance criteria (X-RAY & N-RAY)

1. Verify presence and proper assembly of all parts.
2. Verify explosive charges are free of voids, cracks, and foreign material.
3. No potting compound allowed on spark gap side of gasket seal. (N-ray only)
4. Gasket seal seated properly against body at spark gap. (N-ray only)
5. 0.085 inch max voids allowed ½" from spark gap gasket seal. (N-ray only)
6. 0.160 inch max void allowed in remainder of potting. (N-ray only)

3.4.3 Insulation resistance. Each cartridge's insulation resistance shall be 2 megaohms minimum. The application of insulation resistance voltage shall not dud, adversely affect or cause the cartridge to fire when tested in accordance with 4.4.3.

3.4.4 Bridge circuit resistance. All cartridge's bridge circuit resistance shall be 1.0 ± 01 ohms. The measurement of resistance shall not dud, adversely affect or cause the cartridge to fire when tested in accordance with 4.4.4.

3.4.5 Leakage. Each cartridge's leak rate shall not exceed 1.0 $x10^{-6}$ STD cc/sec of air when tested in accordance with 4.4.5.

3.4.6 Electrostatic discharge. Each cartridge shall be tested in accordance with 4.4.6 and shall not fire or dud.

3.4.7 Power - current. The cartridge shall not fire, be degraded or damage during the No-Fire current application test, see 4.6.

3.5 Ballistic requirements. The cartridges subjected for ballistic testing shall meet the requirements listed below when subjected to the test prescribed in 4.7.

3.5.1 Ignition time. The Maximum ignition time shall be 20 milliseconds, see 4.7.3.

3.5.2 Indent depth. The minimum indent depth shall be 0.023 inches, see 4.7.4.

3.5.3 Misfires. There shall be no misfires, see 4.7.5.

4.0 VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

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4.2 First article inspection. Before entering into quantity production, an acceptable first article sample of 24 cartridges shall be prepared using the same procedures and methods proposed for the production lot. Twenty of these cartridges shall be expended in the tests listed in Table III and four shall be retained for investigative purposes. A first article sample acceptable for functional and environmental testing shall be defined as a sample, which has met the requirements of paragraph 3.2 through 3.4.6 of this specification. The contractor shall deliver the cartridges to the activity for tests designated by the contracting agency. Failure of one or more cartridges to pass the tests shall result in the rejection of the first article. Quantity production prior to notification by the contracting agency of pre-production sample acceptability shall be at the contractor's risk. The pre-production cartridges will not be applied as part of the quantity specified for delivery by the contract.

4.3 Conformance inspection. Samples for ballistic test shall be selected from the production lot in accordance with requirements of Table II. Reduced sample size is recommended after 10 consecutive acceptable lots. Sample plan to be used will be normal unless otherwise specified in the contract. Four additional cartridges shall be provided with each test sample for investigative purposes. Sample cartridges made available for investigative purposes shall not be applied as part of the quantity specified for delivery by the contract or order. The production lot size shall be 500 maximum. The contracting officer shall approve exceptions to lot size. The test samples shall be subjected to the nondestructive tests of 4.4, 4.6 and ballistically tested in accordance with 3.5. The sample of cartridges shall be divided into three equal groups and tested at each of the three temperatures listed in 4.7.1. If the sample cannot be divided equally the additional cartridges will be tested at cold temperature.

Table II. Conformance sample size

Lot Size	Normal Sample Size	Reduced Sample Size
2-8	2	2
9-15	3	2
16-25	5	3
26-50	8	5
51-90	13	5
91-150	20	8
151-280	32	13
281-500	50	20

4.4 Nondestructive tests.

4.4.1 Visual inspections. The external condition and appearance of the cartridges shall be determined by comparison to drawings 3205AS277-1, 3205AS277-2 and 3205AS277-3 as required. All cartridge assemblies shall be visually examined and compared to the appropriate drawing for proper assembly, identification, color and interface control points. Those with illegible or inadequate descriptive markings or serial numbers, dents, deep scratches, or any other defects, which may prevent proper installation or functioning, shall be rejected. Any visual defects found during nondestructive inspection and destructive tests will result in 100 percent screening of the production lot at the contractor's expense. Each cartridge shall meet the requirements of 3.4.1.

4.4.2 Radiographic examination. All assembled cartridges shall be radiographically examined via both X-ray and N-ray. X-ray radiographic examination shall be conducted in accordance with ASTM E1742/E1742M; N-ray examination shall be in accordance with ASTM E 748. The cartridges shall be positioned for the most revealing exposure with the long axis perpendicular to the rays of the radiographic machine. All cartridges shall be identified with serial numbers prior to examination beginning with 001 for

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each production lot. The cartridges shall be arranged on boards or trays in consecutive order, with any missing serial numbers identified on the radiographic plate. Each radiograph shall carry a permanent identification of the items displayed thereon in a 4x6-inch region maximum. The radiographic identification shall include the drawing number, the complete lot number in accordance with MIL-STD-1168, the contract number, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the lot acceptance tests. Any observable imperfections as outlined in 3.4.2 shall be cause for rejection of the cartridge. Defective cartridges found during radiographic review are to be marked on the radiographic plate and removed from the production lot. Settings on the radiographic equipment shall be noted on the radiographic review sheets.

4.4.3 Insulation resistance. The insulation resistance examination shall be conducted in accordance with MIL-STD-202, method 302-test condition B. The 500 VDC potential shall be applied for a minimum of two minutes between the shorted contact pins and the body assembly. Each cartridge shall meet the requirements of 3.4.3.

TABLE III. First article inspection matrix

TEST	GROUP	A	B	C	D	E	F	TOTAL
	QUANTITY	4	4	4	4	4	4	24
	Para #.							
VISUAL EXAMINATION	3.4.1	X	X	X	X	X		20
X-RAY INSPECTION	3.4.2	X	X	X	X	X		20
BRIDGEWIRE RESISTANCE	3.4.4	X	X	X	X	X		20
INSULATION RESISTANCE	3.4.3	X	X	X	X	X		20
BRIDGEWIRE RESISTANCE	3.4.4	X	X	X	X	X		20
LEAKAGE	3.4.5	X	X	X	X	X		20
BRIDGEWIRE RESISTANCE	3.4.4	X	X	X	X	X		20
ESD 25KV	3.4.6	X	X	X	X	X		20
BRIDGEWIRE RESISTANCE	3.4.4	X	X	X	X	X		20
POWER CURRENT	3.4.7	X	X	X	X	X		20
BRIDGEWIRE RESISTANCE	3.4.4	X	X	X	X	X		20
SHOCK	4.5.3			X				4
VIBRATION	4.5.2		X					4
THSA	4.5.1	X						4
VISUAL INSPECTION	3.4.1	X	X	X				12
X-RAY INSPECTION	3.4.2	X	X	X				12
BRIDGEWIRE RESISTANCE	3.4.4	X	X	X				12
LEAKAGE	3.4.5	X	X	X				12
POWER CURRENT	3.4.7	X	X	X				12
BRIDGEWIRE RESISTANCE	3.4.4	X	X	X				12
PERFORMANCE TEST FIRING	3.5							
FIRE @-65°F DENT TEST					4			4
FIRE @70°F DENT TEST		4	4	4				12
FIRE @ 300°F DENT TEST						4		4
RETAINS FOR INVESTIGATION							4	4

4.4.4 Bridge circuit resistance. The resistance of each bridge circuit shall be measured with accuracy of one percent using a test which subjects the bridge circuit to a current of less than 20 milliampere. The test shall be conducted in accordance with MIL-STD-202, method 303 except that the measurements shall be made at 70°F. Each cartridge shall comply with 3.4.4.

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4.4.5 Leakage test. Each cartridge shall be leak tested to determine conformance to 3.4.5. Production units shall be leak tested by flushing the loaded cartridge assembly with Helium at a pressure of 1+0.1/-0.0 atmospheres absolute. The cartridge assembly shall then be washed with dry Nitrogen and placed in a vacuum chamber and checked for leakage at one atmosphere differential. Certification of the test results shall be supplied and Government witnessing will be as specified in the contract or purchase order.

4.4.6 Electrostatic discharge. Each cartridge shall be subjected to an electrostatic discharge of 25,000 ± 500 VDC from a 500 ± 25 Picofarad capacitor through a 5,000 ± 250 ohm resistor connected between shorted bridgewire pins and case.

4.5 Environmental test.

4.5.1 Temperature-shock/humidity/altitude (TSHA). A TSHA test per MIL-DTL-23659, Temperature-Shock/Humidity/Altitude, shall be conducted with the following exceptions.

- **a.** Altitude pressure 80,000 versus 70, 000 feet
- **b.** 95% versus 50% relative humidity (RH) applied at 70 °F
- **c.** 300 °F versus 160 °F conditioning along with omission of RH
- **d.** Place units on screen tray or in wire baskets to allow air to circulate freely, exposing all surfaces of the units, except noted otherwise.

TABLE IV. TSH&A CYCLING PROCEDURE

DAY	TIME	PROCEDURE
MONDAY	0800	Place items in chamber maintained +300 °F
	1200	Chamber to +70°F at 95% RH. Chamber shall reach conditions within one hour.
	1600	Remove items and immediately place in a chamber maintained at –65 °F at an altitude pressure of 80,000 feet (ft)
TUESDAY	0800	Remove items from chamber and place in chamber maintained at +70 °F and 95% RH.
	1200	Remove items from chamber and place in chamber maintained at –65 °F at an altitude pressure of 80,000 ft
	1600	Remove items from chamber and place in chamber maintained at +300 °F.
WEDNESDAY	0800	Reduce chamber pressure to 70 °F at 95% RH. Chamber shall reach conditions within one hour.
	1200	Raise chamber to +300 °F. Chamber shall reach conditions in one hour.
	1600	Remove items and place in chamber maintained at –65 °F at an altitude pressure of 80,000 ft.
THURSDAY		Repeat Tuesday schedule.
FRIDAY	0800	Reduce chamber pressure to +70 °F at 95% RH. Chamber shall reach conditions within one hour.
	1200	Raise chamber to 300 °F. Chamber shall reach conditions in one hour.
	1600	Remove items and place in chamber maintained at +70 °F at 95% RH. See Note (2)

Table notes:

1. Tests shall be performed in the sequence shown, for duration and conditions shown. Day and clock times are provided as a guide and are not mandatory.

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2. Test schedule shall repeated for a total of four weeks (28 days) except that on the second and fourth weekends the soak time was from 1600 Friday until 0800 on Monday shall be at 300°F.

3. No interim withdrawals of items. All items submitted for test shall complete the entire 28-day test cycle.

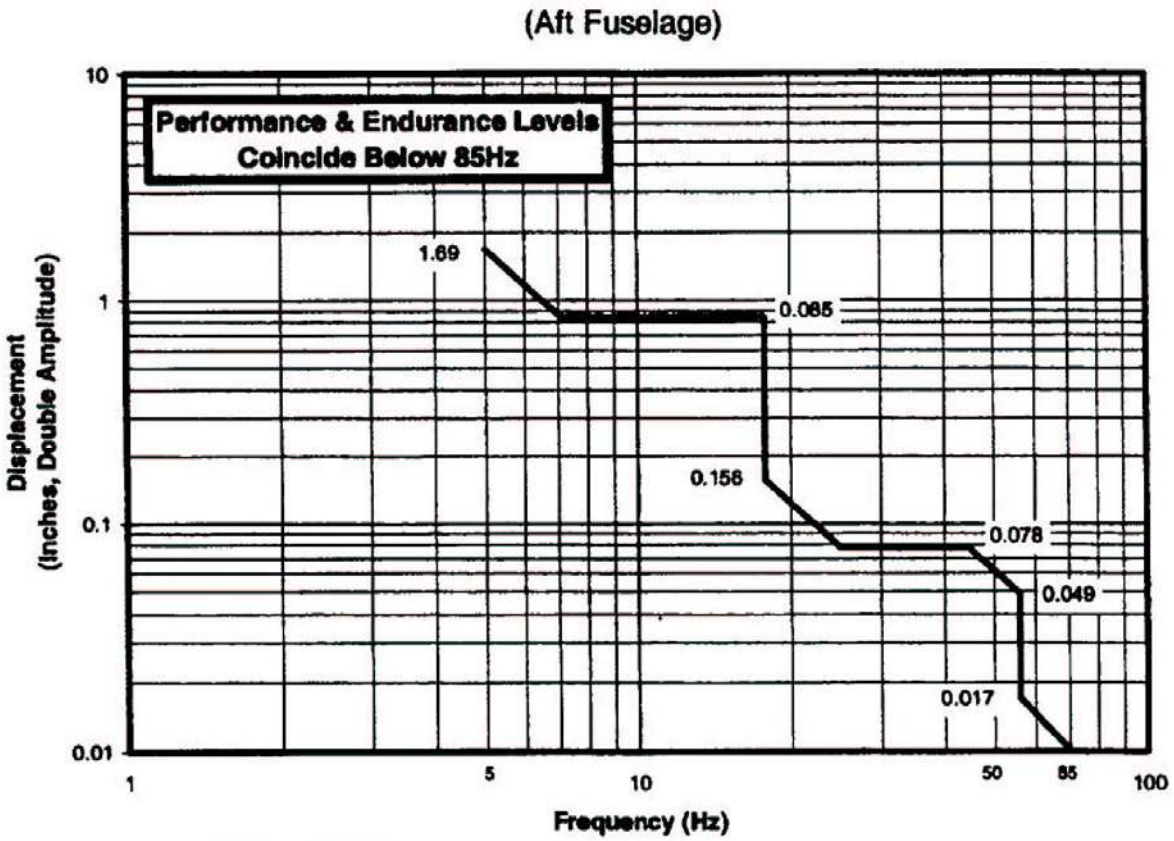
4.5.1.1 Success criteria. There shall be no evidence of excessive corrosion, peeling, chipping or blistering, of the finish or exposure of base metal. The cartridges shall perform satisfactorily when subjected to subsequent testing per Table III.

4.5.2 Vibration (Non-gunfire). Cartridges without protective caps but with shunt plugs shall be vibrated in three mutually perpendicular axes. One axis shall be the longitudinal axis of the cartridge. Each cartridge will be mounted by its threads to the vibration test fixture. The following sinusoidal and random vibration tests shall be conducted at -65°F, +70°F, and +300°F.

a. Sinusoidal Vibration. In each axis and at each temperature the cartridge shall be subjected to 30 minutes of frequency cycling from 5 Hz to 85 HZ and back to5 Hz in 10 minute periods. Total sinusoidal vibration time in each axis will be 90 minutes. The rate of frequency change shall be logarithmic with time. The double amplitude or vibratory acceleration levels will be in accordance with Figure 1.

b. Random Vibration. The initiators shall be vibrated along the same orthogonal axis and temperature as in 4.5.2.a. Cartridges will be vibrated for 30 minutes in each axis at each temperature for a total random vibration time of 270 minutes per axis. Test levels will be in accordance with figure 2 for frequency range of 85 Hz to 2000 Hz only. Random levels shown on figure2 below 85 HZ are not to be used. Endurance overall for 85 to 2000 Hz is 34.82 Grms.

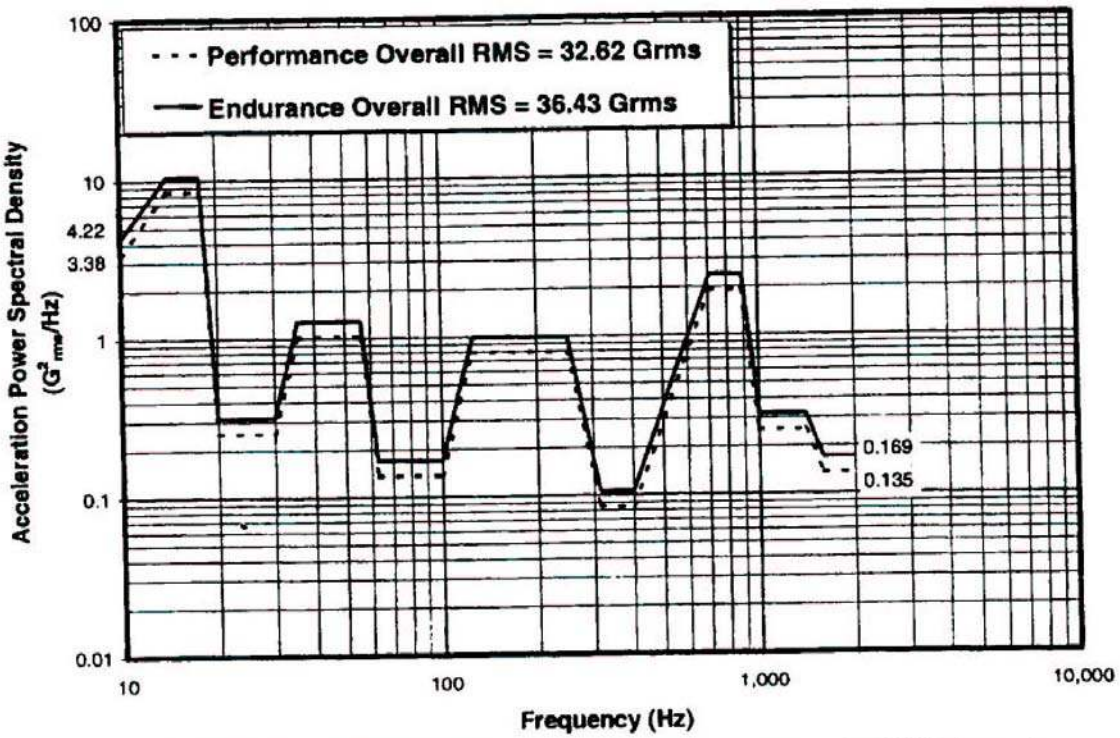
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Non-Gunfire Vibration Region V-AF-1 Sinusoidal	
Frequency Range (Hz)	Displacement or Acceleration
5-7.1	±2.2 G
7.1-17.8	0.845 in D.A.
17.8-25	±2.5 G
25-44.7	0.078 in D.A.
44.7-56.2	±7.9 G
56.2-70.8	±2.7 G
70.8-85	0.01 in D.A.

FIGURE 1. Non gunfire sinusoidal low frequency test levels

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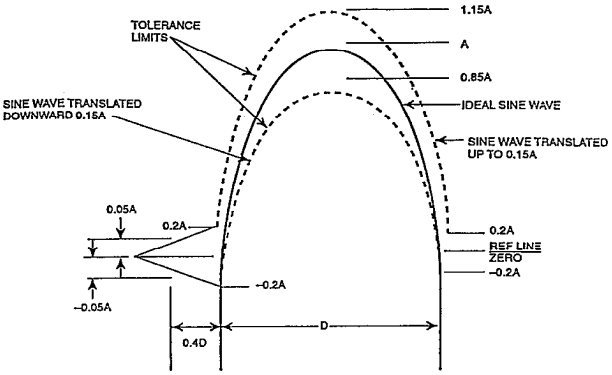
Non-Gunfire Vibration Region V-AF-1 Random			
Performance: Overall 32.62 G_{rms}		**Endurance: Overall 36.43 G_{rms}**	
Frequency Range (Hz)	G^2_{rms}/Hz or dB/Octave	Frequency Range (Hz)	G^2_{rms}/Hz or dB/Octave
10-14.1	8.0 dB/Oct	10-14.1	8.0 dB/Oct
14.1-17.8	8.45 G^2_{rms}/Hz	14.1-17.8	10.54 G^2_{rms}/Hz
17.8-20	-90 dB/Oct	17.8-20	-90 dB/Oct
20-30	0.254 G^2_{rms}/Hz	20-30	0.316 G^2_{rms}/Hz
30-35.5	25 dB/Oct	30-35.5	25 dB/Oct
35.5-56.2	1.014 G^2_{rms}/Hz	35.5-56.2	1.265 G^2_{rms}/Hz
56.2-63	-53 dB/Oct	56.2-63	-53 dB/Oct
63-100	0.135 G^2_{rms}/Hz	63-100	0.169 G^2_{rms}/Hz
100-125	24 dB/Oct	100-125	24 dB/Oct
125-250	0.794 G^2_{rms}/Hz	125-250	0.991 G^2_{rms}/Hz
250-315	-29 dB/Oct	250-315	-29 dB/Oct
315-400	0.085 G^2_{rms}/Hz	315-400	0.105 G^2_{rms}/Hz
400-708	16.5 dB/Oct	400-708	16.5 dB/Oct
708-891	1.91 G^2_{rms}/Hz	708-891	2.38 G^2_{rms}/Hz
891-1000	-53 dB/Oct	891-1000	-53 dB/Oct
1000-1410	0.254 G^2_{rms}/Hz	1000-1410	0.316 G^2_{rms}/Hz
1410-1600	-15 dB/Oct	1410-1600	-15 dB/Oct
1600-2000	0.135 G^2_{rms}/Hz	1600-2000	0.169 G^2_{rms}/Hz

FIGURE 2. Random vibration levels

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4.5.2.1 Success criteria. The cartridges shall perform satisfactorily when subjected to subsequent testing per Table III.

4.5.3 Shock. The cartridges shall be subjected to the 20g-shock test profile outline in Figure 3 while mounted in the shock fixture, figure 2. Apply impact shocks, three in each direction per orthogonal axis depicted in Figure 2.



SERVICE SHOCK Basic Test	
Peak Value (G's)	Pulse Duration (ms)
20	11 ± 1

The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The acceleration amplitude of the ideal half sine pulse is A and its duration is D. Any acceleration pulse, which can be contained between the broken line boundaries when measured with a 250Hz low pass filter, is acceptable.

FIGURE 3. Shock profile.

4.5.3.1 Success criteria. The test items shall not fire; incur damage or degradation during the shock testing. The test items shall not have any loose, broken, or missing components. All test items shall be safe to handle after shock testing. The test items shall not be damaged to the extent that would prevent proper installation. The cartridges shall perform satisfactorily when subjected to subsequent testing per Table III.

4.6 No-Fire test. A direct current of not less than one ampere supplying a minimum of one watt shall be applied to the bridge circuit for a period of at lease 5 minutes. The test current shall be regulated throughout the period of application to within 2 percent. The cartridge shall be conditioned and tested at ambient temperature.

4.6.1 Success criteria. The cartridge shall not fire or incur degradation or damage during the no-fire current application test.

4.7 Ballistic test. The cartridges subjected to ballistic test shall meet the requirements of 3.5.

4.7.1 Ballistic test. Cartridges allocated for ballistic test firing at -65°F ±5°F and 300°F ±5°F shall be temperature conditioned for a period of time neither less than 4 hours not greater than 24 hours. Cartridges allocated for ballistic test firing at +70°F ±5°F shall be temperature conditioned for a period of time not less than 4 hours. If the cartridge is not fired within three minutes after removal from the temperature-conditioning box, it shall be reconditioned. Apply a minimum quantity of silicone compound SAE AS8660 to face of witness block in contact with output end of cartridge. The quick release pin shall be adjusted so that the spring applies a force of 1.6±02 pounds to ensure intimate contact of the block to the cartridge output end. The test specimen shall be fired in a manner as depicted on NAVAIR drawing no. 838AS185. A direct current twenty (20) millisecond pulse shall be applied to the bridge circuit. The test current shall be a DC current pulse, regulated throughout the period of application, to within 2 percent of the desired value. If

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a rectified current is used, the ripple content shall not exceed 5 percent RMS of the test current. The test sample shall be fired at 3.1 amperes and 28 volts DC.

4.7.2 Acceptance Criteria. Acceptance criteria 4.7.1.2 through 4.7.1.6 shall be measured and recorded.

4.7.3 Ignition Time. The ignition time is defined as the time from the first application of firing current until the circuit opens as recorded on the oscillograph trace. The lot shall be acceptable with respect to ignition time if the ballistic sample meets the requirements of 3.5.1.

4.7.4 Indent Depth. The lot shall be acceptable with respect to indent depth if the ballistic sample meets the requirements of 3.5.2.

4.7.5 Misfire. A cartridge that does not fire under the conditions specified in 4.7.1 shall be considered defective, and the lot considered unacceptable, unless the misfire is clearly attributable to faulty test equipment or procedure.

4.7.6 Retest. There shall be no retest.

4.7.6.1 Test Failure. If a test failure is attributable to an assignable cause, excluding the test cartridges, the original test results shall be discarded and the test reconducted.

5.0 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6.0 NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended Use. These aircraft fire extinguisher cartridges are intended for use in the fire extinguisher systems of the F-18A/B/C/D/E/F & AH-1Z aircraft. These cartridges were specifically designed for use in these military aircraft, and thus have no commercial application.

6.2 Acquisition Requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2 and 2.3).

c. Whether a first article inspection is required and. if so, specify the test activity (See 3.1.)

d. List of data required for each First article and production lot.

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e. Direction for shipping radiographic plates of the entire production lot along with ballistic sample to the activity conducting production lot acceptance tests.

f. Markings if other than as specified on drawing 3205AS277 and 520-174-0282.

g. Production lot sizes (quantity), sample size and test facility (See 4.2).

h. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as a part of the description of work to be accomplished by a Government activity the safety precautions requirements of "Ammunition and Explosives," OP 5, are applicable.

6.3 First Article. When a first article inspection is required (see 3.1), the contracting officer should provide specific guidance to offeror(s) whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents of the first articles. Invitation(s) for bid(s) should provide that the U.S. Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the U. S. Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior U.S. Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.3.1 Bar Coding. Bar Coding should be as specified in Section-D of the contract or purchase order (see 6.2), the outer container should be bar coded in accordance with MIL-HDBK-129. Additionally marking information specified on drawing 3205AS277 should be applied to the containers in accordance with the provisions of MIL-HDBK-129.

6.3.2 Explanation of terms. For the purpose of this specification, NAVSUPINST 4030.28B should be used to clarify terms.

6.4 Subject term (keyword) listing.

F-18A/B/C/D/E/F
AH-1Z

6.5 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes. The changes in this revision were authorized by ECP 055220V016.

Preparing activity
Navy - OS
(Project 1377-2011-016)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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